

Lab Goal : This lab was designed to teach you more about graphs, mazes, and recursion.

Lab Description : A maze is a two dimensional structure. All of the mazes will have the entrance point at the top left corner. The exit point will be the bottom right corner.

Find the shortest path through the maze to get from the entrance point to the exit point.

Helpful Hints / Assumptions: All input characters will be '#'s and '.'s, representing walls and open spaces respectively. You can travel through open spaces '.'s but not walls '#'s.

Sample Data :

```
5
..###
...#
###.
#...
###.
9
..#####
...#...#
#...###
#.....#
#...#.#
#.#...#.#
#.#...#.#
#.#...#
#####..
9
..#####
...#...#
#...###
#.....#
#...#.#
#.#...#.#
#.#...#.#
#.#...#
#####..
```

Use your files from the 1st maze lab.

```
//OOP Lab - recursion
Maze.java
MazeRunner.java
```

```
//OOP Lab – queue
MazeQ.java
MazeQRunner.java
```

```
//non-OOP Lab
Maze_Simple.java
```

```
//data file needed for all
maze_simple.dat
```

Sample Output :

```
..###
...#
###.
#...
###.
8 steps
..#####
...#...#
#...###
#.....#
#...#.#
#.#...#.#
#.#...#.#
#.#...#
#####..
16 steps
..#####
...#...#
#...###
#.....#
#...#.#
#.#...#.#
#.#...#.#
#.#...#
#####..
no exit found
```